Pre-requisites

OptiCrop: Smart Agricultural Production Optimization Engine

This document lists all the tools, environments, and libraries used across the project, along with what each one is used for.

Development Environment

Tool	Purpose	Link
Anaconda Navigator	Desktop GUI used to create and manage isolated Python environments and install data-science packages without dependency conflicts.	anaconda.com/download
PyCharm	IDE used to write, organize, and debug the Python codebase (data processing, model training, and Flask app).	jetbrains.com/pycharm

Python Libraries

Library	Used In	Purpose
NumPy	Pre-processing, Model Building	Core numerical computing — array operations, mathematical functions, and handling feature matrices for the model.
Pandas	Data Collection, EDA, Pre-processing	Loading the crop dataset (CSV), cleaning data, handling missing values/duplicates, and feature engineering.
Matplotlib	Data Collection & Analysis (EDA)	Base plotting library used for univariate distribution plots (histograms) and general charting.
Seaborn	Data Collection & Analysis (EDA)	Statistical visualizations built on Matplotlib — boxplots (bivariate analysis), correlation heatmaps and pairplots (multivariate analysis).
Scikit-learn	Pre-processing, Model Building	Train/test splitting, feature scaling, K-Means Clustering (grouping similar agricultural conditions), Logistic Regression (crop classification), and evaluation metrics.
Flask	Application Building	Lightweight web framework used to build the backend that loads the trained model and serves crop predictions to the front end.

Front-End

Tool	Purpose
HTML/CSS	Used to design the input form and results page where users enter soil/climate parameters (N, P, K, temperature, humidity, pH, rainfall) and view the recommended crop.

Model Persistence

Tool	Purpose
Pickle (Python standard library)	Used to save the trained model and scaler objects so they can be loaded directly by the Flask app without retraining.

Dataset

Crop Recommendation Dataset — contains 2,200 records across 22 crop types, with 7 features: N, P, K, temperature, humidity, ph, rainfall, and target label (crop name).